Prophet Model Implementation

UPI Forecasting

With holiday and event effects



Group 2

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Why Forecast UPI?

- Daily volume swings with the calendar
- Festivals, paydays and sales cause spikes
- Banks plan capacity from daily forecasts

+ 790M

per day, Aug 2026

+ 5.3x

growth since Jan 2022



Our Questions

+1 Do holidays and events improve forecasts?

+2 Does Prophet recover the true effects?

+3 Does Prophet beat SARIMAX?

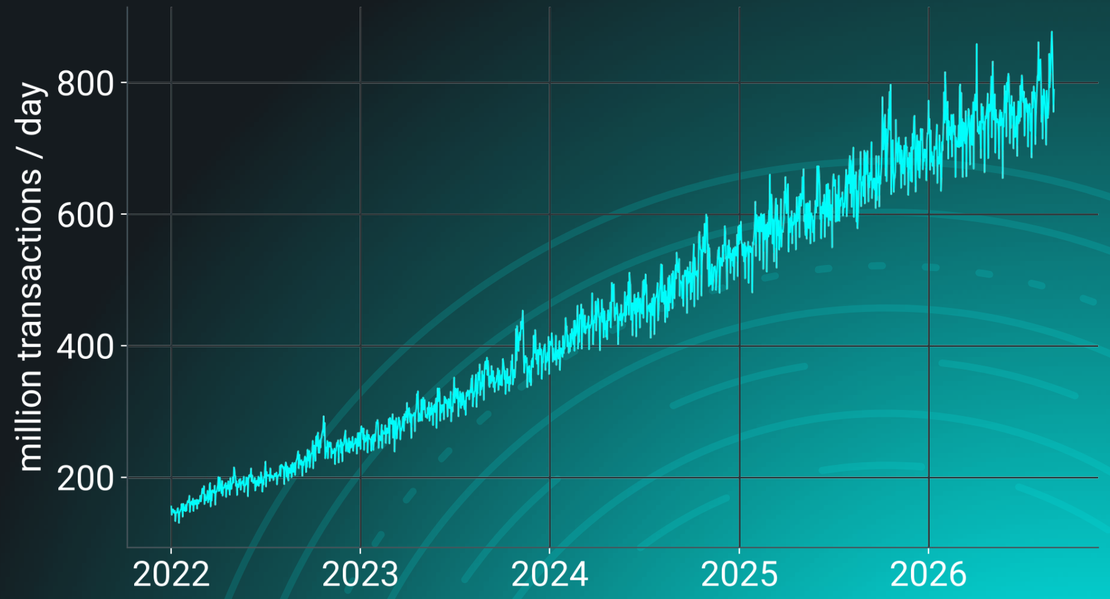
+4 Forecast volume to February 2027

Our Data

- Real NPCI monthly UPI totals
- January 2022 to August 2026
- NPCI publishes no daily figures
- So we built a daily series
- Known effects let us check accuracy

56
months of data

1,704
daily values



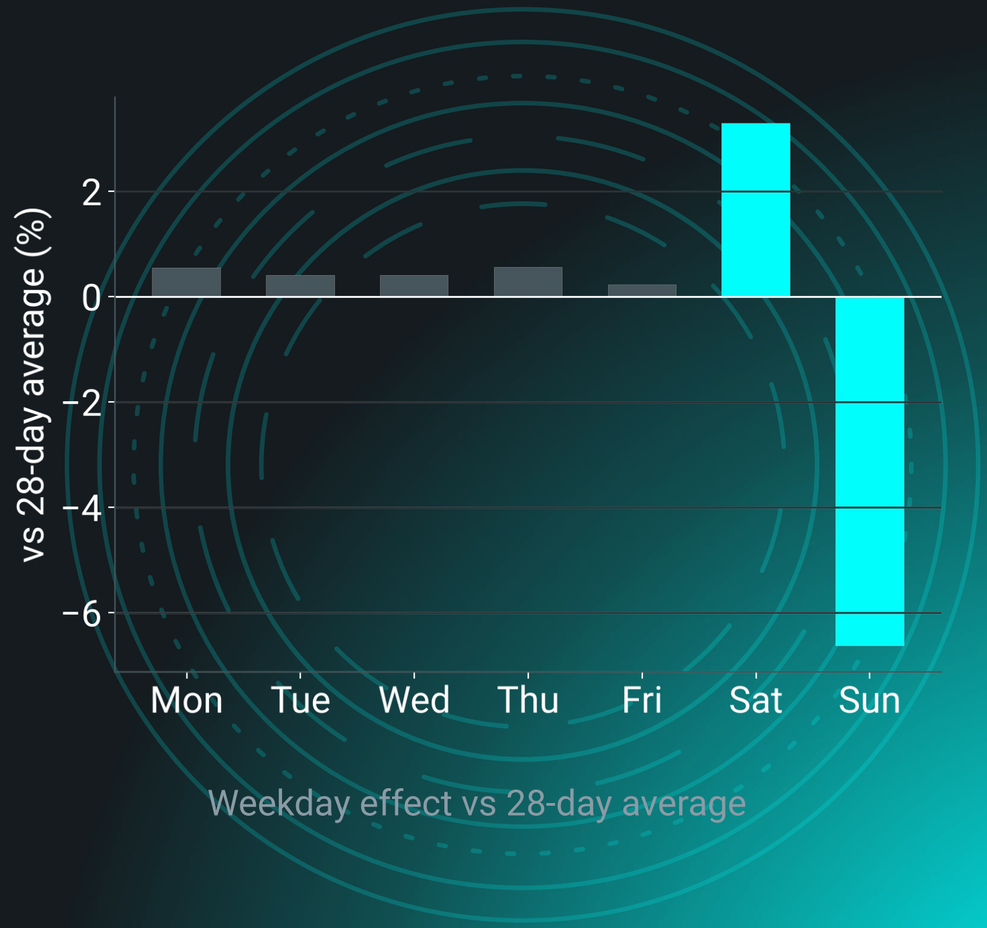
Holidays and Events

- Festivals: Diwali, Holi, Eid, Raksha Bandhan
- National holidays as single days
- Events: sale, payday and month-end



What the Data Shows

- Volume grew five times since 2022
- Saturday highest, Sunday about 7% lower
- Paydays lift volume about 4.5%
- Festivals add bumps each October
- One difference makes it stationary



Our Models

- Baseline: same weekday last week



Prophet

Trend and seasonality

Plus holiday windows

Plus payday and sale

SARIMAX

Order chosen by AIC

Same calendar inputs

How We Tested

- Rolling origin cross validation
- Same 10 folds for every model
- Scored by MAPE, MAE and RMSE



TRAIN ON TWO YEARS

FORECAST 90 DAYS

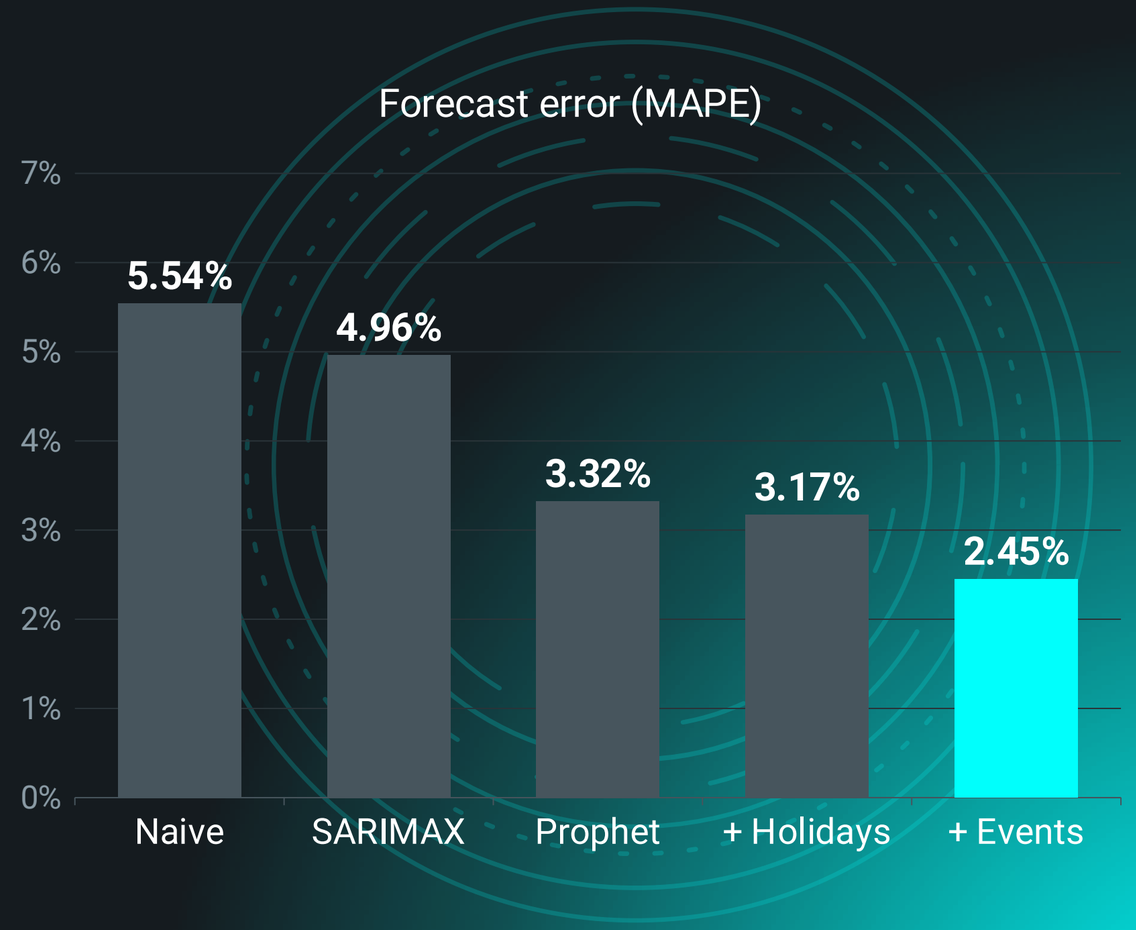
ROLL FORWARD, REPEAT

Results at a Glance

- Prophet with holidays and events best
- SARIMAX trails every Prophet version

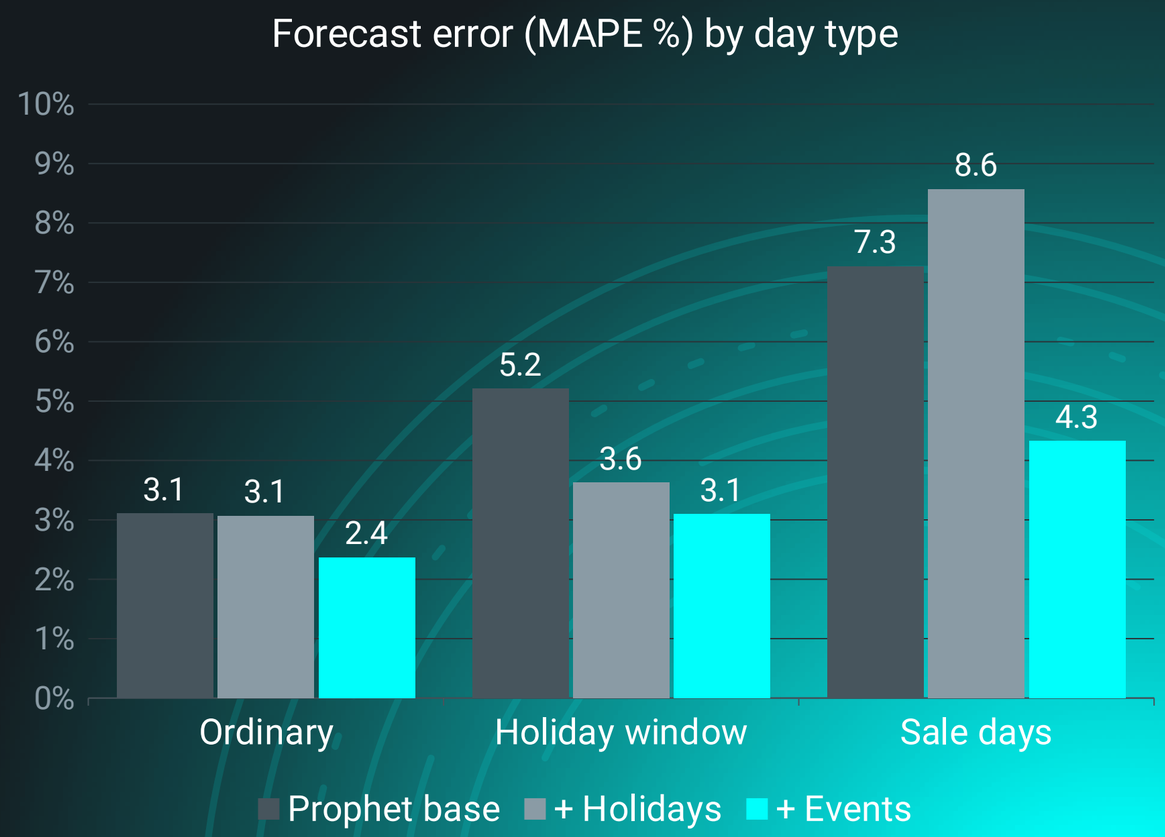
+ 2.45%
best model error

+ 5.54%
naive baseline error



Where Error Drops

- Holidays cut holiday-day error 30%
- Events halve error on sale days
- Paydays help ordinary days too



Estimated vs True

Estimated +6.2%
True +6.0%

Payday
Almost exact

Estimated +9.8%
True +10.7%

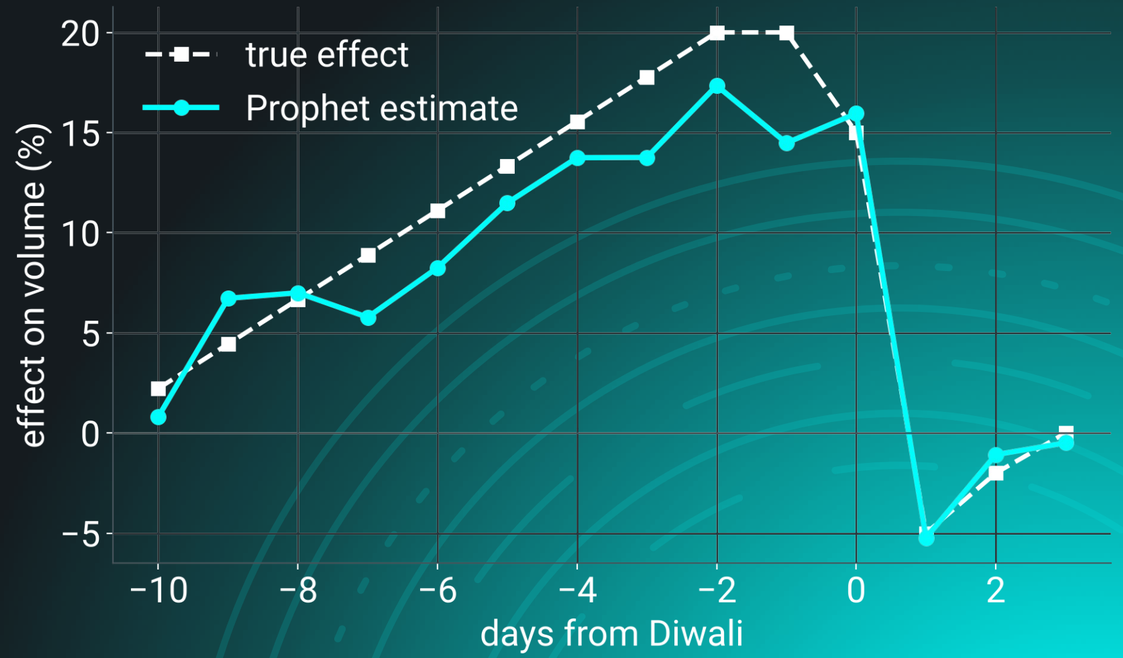
E-commerce sale
Close

Estimated +7.7%
True +9.1%

Diwali window
Underestimated

The Diwali Effect

- Ten-day ramp captured well
- Peak estimated 17%, true 20%
- Only four Diwalis to learn from
- Yearly cycle absorbs some effect



Model Checks

Why did SARIMAX lose? −

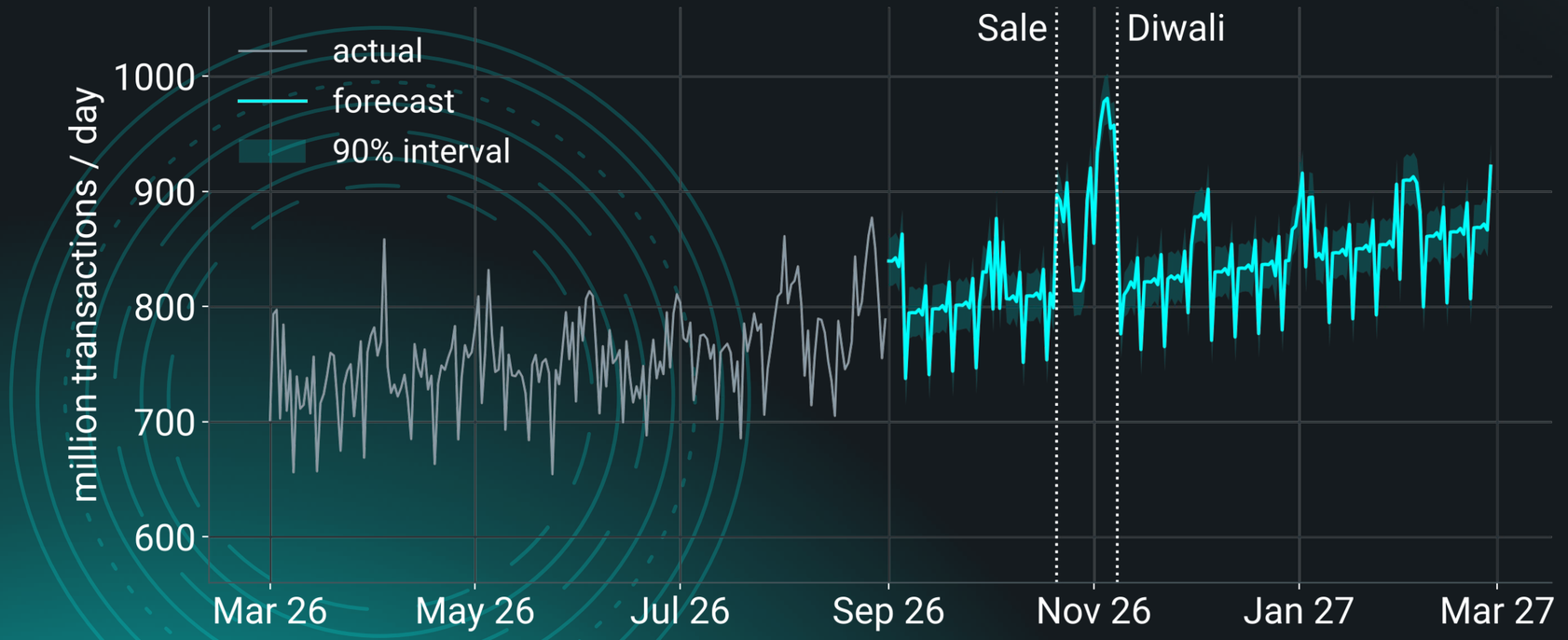
- Its errors grow over 90 days
- It overshoots holiday days by 6%

Prophet errors still correlated +

Error size changes over time +



Six Month Forecast



- 24 to 26 billion a month
- Sale and Diwali peaks visible
- Real range likely wider

Conclusions

+1 Holidays and events improve accuracy

+2 Prophet recovers most effects closely

+3 Prophet beats SARIMAX out of sample

+4 Diwali size still underestimated

Looking Ahead



Limitations

Synthetic daily data

Only four Diwalis

Intervals too narrow

Future Work

Real daily data

Prophet plus ARIMA

Compare with LSTM

Thank You

Questions are welcome



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